

MiroFish Community

Community Context Document

Urban Flood Evacuation Scenario — Synthetic Case Study

Purpose: This document is designed as seed/context material for MiroFish Community. It provides a structured fictional community, stakeholder network, emergency scenario, alternative decisions, behavioural conditions and research variables for multi-agent social decision simulation.

Important: All population figures, stakeholder descriptions, relationships, events and conditions in this document are synthetic assumptions created for research experimentation. They do not represent actual data about any real Mumbai community.

1. Synthetic Community Profile

Attribute	Synthetic Description
Community	Harbour View Community
Setting	Dense fictional urban neighbourhood in the Mumbai metropolitan context
Population	Approximately 8,000 fictional residents
Housing	Apartment buildings, older low-rise buildings and rental housing
Economy	Small shops, local services, informal work and office commuters
Mobility	Walking, buses, auto-rickshaws, private vehicles and two-wheelers
Facilities	School, clinic, community hall, market and local transit points
Main hazard	Rapid urban flooding following intense rainfall
Emergency challenge	Limited time, transport, shelter capacity and emergency resources

2. Flood Scenario and Timeline

Harbour View Community receives a warning that intense rainfall is expected to continue. Water begins accumulating on several low-lying roads. As conditions deteriorate, residents must decide whether to evacuate, wait for additional information, assist neighbours, protect property or seek alternative routes.

Stage	Synthetic Event
T-3 hours	Weather conditions worsen. Informal messages begin circulating among residents.
T-2 hours	Local authorities issue an initial flood-risk warning.
T-90 minutes	Low-lying roads begin experiencing significant water accumulation.
T-60 minutes	Emergency responders recommend preparing vulnerable residents for evacuation.
T-30 minutes	Some transport routes become unreliable. Conflicting informal messages appear.
T-0	Authorities must select and communicate an evacuation strategy.
T+30 minutes	Residents react to official instructions and observations from neighbours.
T+60 minutes	Shelter demand, transport availability and information diffusion become important.
T+120 minutes	Community-level outcomes emerge from individual decisions and interactions.

3. Stakeholder Groups

The virtual community should not behave as a homogeneous population. Agents should represent different priorities, constraints, beliefs, relationships and levels of influence.

General residents: Prioritize personal and household safety, reliable information and convenient evacuation.

Parents and caregivers: Prioritize children and dependents; may delay evacuation until family members are together.

Elderly residents: May have reduced mobility and greater dependence on assistance or accessible transport.

Vulnerable / medically dependent residents: May require medication, medical support, accessible transport and appropriate shelter.

Students and young adults: Generally more mobile and potentially active in peer-to-peer information sharing.

Shopkeepers and small businesses: May worry about inventory, property security, income loss and evacuation timing.

Daily-wage / informal workers: May face stronger economic pressure to continue working despite rising risk.

Emergency responders: Prioritize rescue, route clearance, vulnerable households and efficient resource allocation.

Local officials: Coordinate warnings, evacuation instructions, shelters, transport and public communication.

Healthcare workers: Prioritize patients, medical supplies and continuity of essential services.

Community volunteers: Help distribute information, identify vulnerable households and support evacuation.

Transport operators: Balance passenger safety, route conditions and service availability.

4. Community Relationships and Information Network

Agents are connected through households, neighbourhood ties, workplaces, schools, businesses, community organizations and online communication. These relationships determine how information travels and how individual decisions can influence others.

Family → shared decisions, responsibility and mutual assistance.

Neighbours → local observations, evacuation assistance and informal warnings.

Students / peers → rapid peer-to-peer information diffusion.

Shopkeepers ↔ residents → local information and economic dependencies.

Officials → residents → official warnings and evacuation instructions.

Emergency responders → officials → operational information and resource constraints.

Volunteers → vulnerable residents → assistance and local information.

Healthcare workers → vulnerable residents → medical guidance and support.

Social networks → community → rumours, verification, support and behavioural influence.

5. Trust and Information Conditions

Not all agents trust the same information source. Some residents strongly trust official alerts, while others rely on family members, neighbours, local leaders or direct observation. Information may arrive at different times and may sometimes conflict.

Official messages are generally considered authoritative but may be perceived as delayed.

Neighbour reports may be fast but incomplete.

Family members can strongly influence individual decisions.

Social-media-style messages may spread quickly and can contain both accurate information and rumours.

Agents may seek confirmation before acting.

Visible behaviour of nearby residents can become a signal that influences undecided agents.

6. Flood-Related Constraints

Some roads become difficult or impossible for vehicles to use.

Emergency responders have limited vehicles, personnel and time.

Shelter capacity is smaller than the total community population.

Transport availability decreases as flood conditions worsen.

Some households contain members who cannot evacuate independently.

Residents may be reluctant to leave property unattended.

Businesses may resist early evacuation because of economic losses.

Communication quality differs between highly connected and isolated residents.

Conflicting information may create hesitation, panic or disagreement.

7. Candidate Community Decisions

The same synthetic community should be exposed to alternative decisions so that their possible social consequences can be compared.

Decision	Strategy
A — Immediate Mass Evacuation	Issue a broad evacuation order and encourage residents to leave as quickly as possible.
B — Phased Evacuation	Evacuate the highest-risk areas first, followed progressively by lower-risk areas.
C — Vulnerable-First Evacuation	Prioritize elderly residents, children, people with disabilities and medically vulnerable households.
D — Shelter-in-Place + Targeted Evacuation	Keep residents in safer buildings where appropriate while evacuating households in the highest-risk locations.

8. Possible Behavioural Reactions

Immediate compliance with an official evacuation order.

Delayed compliance while waiting for additional confirmation.

Refusal to evacuate because of property or economic concerns.

Households splitting temporarily because members receive different information.

Residents helping elderly or vulnerable neighbours.

Residents sharing warnings through informal networks.

Rumours increasing uncertainty or resistance.

Groups forming around trusted local information sources.

Conflict between safety priorities and business continuity.

Increased cooperation after residents observe successful evacuation by others.

Changes in perceived risk after direct observation of flooding.

9. Research Questions

Which evacuation strategy produces the highest simulated community acceptance?

Which strategy produces the strongest consensus and lowest polarization?

How rapidly does evacuation information diffuse through the community?

Which stakeholder groups show the greatest support or resistance?

Which agents become influential during the decision process?

How does behaviour change when agents observe the actions of others?

What unintended social consequences emerge under each strategy?

Does prioritizing vulnerable residents affect overall community cooperation?

Are scenario outcomes consistent across repeated stochastic simulation runs?

Can multi-agent simulation provide useful comparative evidence for community decision support?

10. Social Impact Metrics

Metric	What it represents
Acceptance	Degree of support or compliance with a proposed decision.
Consensus	Extent to which agents converge toward a shared position.
Polarization	Separation of agents into opposing behavioural or opinion groups.
Conflict	Frequency or intensity of disagreement and competing priorities.
Information diffusion	How quickly and widely important information spreads.
Influence	Agents or groups that substantially affect others' decisions.
Adoption / compliance	Extent to which agents follow the selected strategy.
Equity	Differences in outcomes experienced by stakeholder groups.
Stability	Consistency of observed outcomes across repeated simulations.

11. Experimental Design

For comparison, the community context and stakeholder-generation process should remain as consistent as possible while the candidate decision changes. Each scenario can be simulated multiple times with different stochastic conditions. Results should be reported as distributions or ranges where appropriate rather than relying on a single run.

Example: Scenario A → repeated simulation runs → social impact metrics → Scenario B → repeated runs → metrics → Scenario C → repeated runs → metrics → Scenario D → repeated runs → metrics → comparative analysis.

12. Expected Simulation Outputs

Stakeholder-level reactions and opinions.

Timeline of major interactions and behavioural changes.

Information diffusion patterns.

Support and resistance patterns.

Emergent cooperation, conflict or polarization.

Influential agents and groups.

Scenario-level social impact indicators.

Comparative analysis of Decisions A, B, C and D.

Decision-support report describing trade-offs, risks and simulated social viability.

13. Interpretation Guidelines

The simulation represents a virtual community, not the actual population. Its outputs should be treated as possible outcomes under specified assumptions. A strategy receiving stronger simulated acceptance does not automatically mean it will succeed in the real world.

The research should distinguish between **simulation outcome**, **comparative scenario result** and **real-world evidence**. Real-world deployment would require empirical data, calibration, validation and human oversight.

14. Seed Entities and Relationships

Entities: Harbour View Community; Residents; Parents; Children; Elderly Residents; Vulnerable Residents; Students; Shopkeepers; Daily-Wage Workers; Emergency Responders; Local Officials; Healthcare Workers; Community Volunteers; Transport Operators; Community Hall; School; Clinic; Market; Evacuation Routes; Shelters; Flood Warning; Evacuation Strategy A; Evacuation Strategy B; Evacuation Strategy C; Evacuation Strategy D.

Key relationships: Residents live in Community; Parents care for Children; Volunteers assist Vulnerable Residents; Officials issue Flood Warnings; Emergency Responders coordinate with Officials; Healthcare Workers support Vulnerable Residents; Shopkeepers depend on Residents; Residents exchange information with Neighbours; Students exchange information with Peers; Transport Operators support Evacuation; Evacuation Strategies affect Residents; Shelters receive Evacuated Residents; Flood Conditions affect Routes and Transport.

15. Synthetic Data Notice

This entire document is fictional and was created specifically for testing and demonstrating the MiroFish Community research framework. The population size, stakeholder characteristics, timeline, infrastructure, relationships, behavioural assumptions and candidate strategies are synthetic. They must not be interpreted as verified statistics, forecasts or recommendations for an actual Mumbai locality.